

# PAPER\_388 — Yang-Mills Mass Gap via SCm Vacuum Density Ratio Evolution

**Source:** grok\_share\_cfdcad2f5.txt, lines ~1-3200 (UQFF Resonance proof set analysis)

**Section:** UQFF\_Resonance Superconductive Universal Gravity Equation system proof set.\_15May2025.docx

**Session:** 106 (grok\_share\_cfdcad2f5.txt full analysis)

**CP4 Class:** YangMillsMassGapVacuumDensityEvolutionCalculator (CP4 #39)

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## Abstract

This paper presents a UQFF analysis of Yang-Mills Mass Gap via SCm Vacuum Density Ratio Evolution, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

## 1. Overview

The Yang-Mills mass gap problem is one of the seven Millennium Prize Problems. It asks for a proof that a Yang-Mills quantum field theory in 4D Minkowski space has a positive mass gap  $\Delta > 0$ .

PAPER\_380 captured a first UQFF Yang-Mills formula using static Meissner-type exponential suppression:

$$\Delta_{\text{PAPER}_380} = \frac{\Phi_{\text{flux}}}{c} \cdot e^{-1}$$

The Star Magic\_construction\_file\_040ct2025.docx thread introduces a **dynamical vacuum-density evolution formula** for the Yang-Mills mass gap that is distinct from PAPER\_380 in: 1. Using time-evolving vacuum density (not static flux) 2. Incorporating the SCm/UA density ratio as a power-law amplifier 3. Employing a double-exponential Gumbel suppression function

This represents the **second distinct UQFF approach** to the Yang-Mills mass gap.

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## 2. The Yang-Mills Mass Gap Equation

### 2.1 Master Formula

$$\Delta m = \sqrt{\dot{\rho}_{\text{vac,UA}} \cdot \left( \frac{\rho_{\text{vac,SCm}}}{\rho_{\text{vac,UA}}} \right)^n \cdot \exp\left(-\exp\left(-\pi - \frac{t}{\text{year}}\right)\right)}$$

Where: -  $\Delta m$  = Yang-Mills mass gap  $(\text{kg} \cdot \text{m}^{-3})^{\{1/2\}}$  or normalized mass unit -  $\dot{\rho}_{\text{vac,UA}} = d\rho_{\text{vac,UA}}/dt$  = time derivative of UA vacuum density -  $\rho_{\text{vac,SCm}}$  = Superconductive medium vacuum density -  $\rho_{\text{vac,UA}}$  = Universal Aether vacuum density -  $n$  = iteration/mode number (integer,  $n \geq 1$ ) -  $t$  = time (s), normalized to 1 year =  $3.156 \times 10^7$  s -  $\pi = 3.14159...$

## 2.2 Component Analysis

### Component 1: Vacuum Density Time Derivative

$$\dot{\rho}_{\text{vac,UA}} = \frac{d\rho_{\text{vac,UA}}}{dt}$$

Using the calibrated UQFF decay law (PAPER\_353):

$$\rho_{\text{vac,UA}}(t) = \rho_{\text{vac,UA}}^{(0)} \cdot \exp(-\exp(-\kappa t))$$

Taking the derivative:

$$\dot{\rho}_{\text{vac,UA}} = \rho_{\text{vac,UA}}^{(0)} \cdot \kappa \cdot \exp(-\kappa t) \cdot \exp(-\exp(-\kappa t))$$

### Component 2: SCm/UA Density Ratio Power Law

$$R_n = \left( \frac{\rho_{\text{vac,SCm}}}{\rho_{\text{vac,UA}}} \right)^n$$

With calibrated values: -  $\rho_{\text{vac,UA}} \approx 6 \times 10^{-27} \text{ kg/m}^3$  (from  $\rho_{\text{vac}} = 6 \times 10^{-27}$  global constant) -  $\rho_{\text{vac,SCm}} \approx f_{\text{SCm}} \cdot \rho_{\text{vac,UA}}$ , where  $f_{\text{SCm}} = 0.001$  (Session 94 calibration)

Therefore:  $\rho_{\text{vac,SCm}}/\rho_{\text{vac,UA}} = 0.001 = 10^{-3}$

For mode  $n$ :  $R_n = (10^{-3})^n = 10^{-3n}$

This power law drives  $\Delta m$  to progressively smaller values for higher modes, consistent with Regge-trajectory-like mass gap scaling.

### Component 3: Double-Exponential Gumbel Suppression

$$G(t) = \exp\left(-\exp\left(-\pi - \frac{t}{\text{year}}\right)\right)$$

This is a **Gumbel/Gompertz** suppression function. Analysis:

| t (years) | Inner exp argument   | G(t)                                                 |
|-----------|----------------------|------------------------------------------------------|
| 0         | $-\pi = -3.1416$     | $\exp(-e^{-\pi}) = \exp(-0.04322) \approx 0.9577$    |
| 1         | $-\pi - 1 = -4.1416$ | $\exp(-e^{-4.1416}) = \exp(-0.01597) \approx 0.9842$ |
| 10        | $-\pi - 10 = -13.14$ | $\exp(-e^{-13.14}) \approx 1 - 2 \times 10^{-6}$     |
| $\infty$  | $-\infty$            | $\exp(0) = 1$                                        |

The suppression is strongest at  $t = 0$ :  $G(0) \approx 0.9577$ , approaching unity asymptotically. At  $t = 0$ , approximately 4.23% suppression is applied to all modes.

The  $\pi$  shift ensures the suppression begins in a physically motivated range — the argument  $-\pi$  places the function at the Gumbel distribution's standard-parameter inflection point.

### 3. Full Evaluation: Mode n=1, t=0

With canonical parameters:  $\rho_{\text{vac,UA}}^{(0)} = 6 \times 10^{-27} \text{ kg/m}^3$  -  $\kappa = 0.0005 \text{ day}^{-1} = 5.787 \times 10^{-9} \text{ s}^{-1}$  - At  $t = 0$ :  $\dot{\rho}_{\text{vac,UA}} = \rho_0 \cdot \kappa \cdot e^0 \cdot e^{-1} = 6 \times 10^{-27} \cdot 5.787 \times 10^{-9} \cdot e^{-1}$

$$\dot{\rho}_{\text{vac,UA}}(0) = 6 \times 10^{-27} \cdot 5.787 \times 10^{-9} \cdot 0.3679 \approx 1.279 \times 10^{-35} \text{ kg/(m}^3 \cdot \text{s)}$$

For n=1:

$$R_1 = 10^{-3}$$

$$G(0) = \exp(-e^{-\pi}) \approx 0.9577$$

$$\Delta m(n = 1, t = 0) = \sqrt{1.279 \times 10^{-35} \cdot 10^{-3} \cdot 0.9577}$$

$$\Delta m(n = 1, t = 0) = \sqrt{1.225 \times 10^{-38}} \approx 3.5 \times 10^{-19} \text{ (kg} \cdot \text{m}^{-3} \cdot \text{s}^{-1})^{1/2}$$

### 4. Distinction from PAPER\_380

| Feature | PAPER_380 (Static Meissner)                  | PAPER_388 (Dynamic Evolution)                                       |
|---------|----------------------------------------------|---------------------------------------------------------------------|
| Formula | $\Delta = \Phi_{\text{flux}}/c \cdot e^{-1}$ | $\Delta m = \sqrt{\dot{\rho}_{\text{vac,UA}} \cdot R_n \cdot G(t)}$ |

| Feature            | PAPER_380 (Static Meissner)             | PAPER_388 (Dynamic Evolution)                  |
|--------------------|-----------------------------------------|------------------------------------------------|
| Time dependence    | Static                                  | Dynamic ( $t$ -dependent)                      |
| Physical mechanism | Meissner-type flux suppression          | Vacuum density ratio evolution                 |
| SCm/UA coupling    | Implicit via $\Phi_{\text{flux}}$       | Explicit power-law ratio                       |
| Suppression type   | Single exponential $e^{-1}$             | Double-exponential Gumbel                      |
| Mode structure     | Single value                            | Infinite mode series in $n$                    |
| Source document    | Master UQFF<br>Resonance_14May2025.docx | Star Magic_construction<br>file_040ct2025.docx |

## 5. Physical Interpretation

The formula captures how the Yang-Mills mass gap arises from:

1. **Vacuum density evolution:** The UA vacuum density decays (PAPER\_353), and its rate of change  $\dot{\rho}_{\text{vac,UA}}$  represents the energy flux driving quantum field excitations.
2. **SCm/UA stratification:** The ratio  $(\rho_{\text{SCm}}/\rho_{\text{UA}})^n$  describes the hierarchical suppression through  $n$  layers of SCm-mediated vacuum — analogous to quantum tunneling through  $n$  barriers, each reducing amplitude by  $10^{-3}$ .
3. **Gumbel temporal suppression:** The double-exponential  $G(t)$  ensures the mass gap is maximally constrained at early times and relaxes toward its asymptotic value as the vacuum stabilizes. This mirrors the Gompertz growth model used in biological and cosmological contexts.
4. **Positivity guarantee:** Since all terms under the square root are positive ( $\dot{\rho} > 0$  for  $t > 0$  and small  $\kappa t$ ;  $R_n > 0$ ;  $G(t) > 0$ ), the mass gap  $\Delta m > 0$  is guaranteed — consistent with the Millennium Prize requirement.

## 6. Mode Spectrum

| Mode $n$ | $R_n$      | $\Delta m$ at $t=0$ (normalized) |
|----------|------------|----------------------------------|
| 1        | $10^{-3}$  | $3.50 \times 10^{-19}$           |
| 2        | $10^{-6}$  | $1.11 \times 10^{-20}$           |
| 3        | $10^{-9}$  | $3.50 \times 10^{-22}$           |
| $n$      | $10^{-3n}$ | $\propto 10^{-3n/2}$             |

The mode spectrum follows  $\Delta m(n) \propto 10^{-3n/2}$ , giving a geometric Regge-like mass ladder with ratio  $10^{-3/2} \approx 0.0316$  between consecutive modes.

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## 7. Validation Cross-Reference

| Reference | Connection                                                          |
|-----------|---------------------------------------------------------------------|
| PAPER_380 | First UQFF Yang-Mills formula (Meissner static) — distinct approach |
| PAPER_353 | Double-exponential vacuum decay law for $\rho_{\text{vac,UA}}(t)$   |
| PAPER_341 | $\kappa=0.0005/\text{day}$ calibration used in density derivative   |
| PAPER_372 | Compressed MUGE SCm/UA density stratification                       |

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**Discovery Class:** Second UQFF Yang-Mills mass gap formula — dynamical vacuum density evolution

**Distinct from:** PAPER\_380 (Meissner static suppression with  $\Phi_{\text{flux}}$ )

**Key feature:** Double-exponential Gumbel suppression + SCm/UA power-law ratio;  $\Delta m > 0$  guaranteed